

MainActivity.kt

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package com.example.notifications

import android.Manifest
import android.app.NotificationChannel
import android.app.NotificationManager
import android.content.Context
import android.content.pm.PackageManager
import android.os.Build
import android.os.Bundle
import android.widget.TextView
import android.widget.Toast
import androidx.appcompat.app.AppCompatActivity
import androidx.core.app.ActivityCompat
import androidx.core.app.NotificationCompat
import androidx.core.content.ContextCompat
import com.google.android.material.button.MaterialButton
import com.google.android.material.textfield.TextInputEditText
import java.io.File

class MainActivity : AppCompatActivity() {

    private lateinit var editText: TextInputEditText
    private lateinit var textView: TextView
    private lateinit var btnSave: MaterialButton
    private lateinit var btnRead: MaterialButton

    private val channelId = "file_channel"
    private val fileName = "user_message.txt"
    private val notificationPermissionRequestCode = 101

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main)

        editText = findViewById(R.id.editText)
        textView = findViewById(R.id.textView)
        btnSave = findViewById(R.id.btnSave)
        btnRead = findViewById(R.id.btnRead)

        btnSave.setOnClickListener {
            val message = editText.text.toString().trim()
            if (message.isNotEmpty()) {
                saveToFile(message)
                editText.text?.clear()
                checkAndShowNotification("Message Saved", "Your message has been stored on the SD card.")
            } else {
                Toast.makeText(this, "Please enter a message", Toast.LENGTH_SHORT).show()
            }
        }

        btnRead.setOnClickListener {
            readFromFile()
        }
    }

    private fun saveToFile(text: String) {
        try {
            val file = File(getExternalFilesDir(null), fileName)
            // Overwrites previous content for this specific request
            file.writeText(text)
            Toast.makeText(this, "Saved Successfully", Toast.LENGTH_SHORT).show()
        } catch (e: Exception) {
            e.printStackTrace()
            Toast.makeText(this, "Save Failed: ${e.message}", Toast.LENGTH_SHORT).show()
        }
    }

    private fun readFromFile() {

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    try {
        val file = File(getExternalFilesDir(null), fileName)
        if (file.exists()) {
            val content = file.readText()
            textView.text = content
            checkAndShowNotification("Message Read", "Content retrieved from the SD card.")
        } else {
            textView.text = "No saved message found."
        }
    } catch (e: Exception) {
        e.printStackTrace()
        textView.text = "Error reading file."
    }
}

private fun checkAndShowNotification(title: String, content: String) {
    if (Build.VERSION.SDK_INT >= Build.VERSION_CODES.TIRAMISU) {
        if (ContextCompat.checkSelfPermission(this, Manifest.permission.POST_NOTIFICATIONS) !=
            PackageManager.PERMISSION_GRANTED) {
            ActivityCompat.requestPermissions(this, arrayOf(Manifest.permission.POST_NOTIFICATIONS),
                notificationPermissionRequestCode)
        } else {
            showNotification(title, content)
        }
    } else {
        showNotification(title, content)
    }
}

private fun showNotification(title: String, content: String) {
    val manager = getSystemService(Context.NOTIFICATION_SERVICE) as NotificationManager

    if (Build.VERSION.SDK_INT >= Build.VERSION_CODES.O) {
        val channel = NotificationChannel(
            channelId,
            "File Operations",
            NotificationManager.IMPORTANCE_DEFAULT
        )
        manager.createNotificationChannel(channel)
    }

    val notification = NotificationCompat.Builder(this, channelId)
        .setSmallIcon(android.R.drawable.ic_dialog_info)
        .setContentTitle(title)
        .setContentText(content)
        .setOngoing(true)
        .setWhen(System.currentTimeMillis())
        .setShowWhen(true)
        .setPriority(NotificationCompat.PRIORITY_DEFAULT)
        .setAutoCancel(false)

    manager.notify(1, notification.build())
}
}

```